MVP7 on the Occurrence–Phase–Atomic Corridor: A Dynamic Source-Locked Route Decision after Metric All-Prefix Return

Liang Wang
School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

TPC-171 integrates the TPC-163–170 batch into a typed source-locked DAG. The structural gap is no longer one opaque crosswalk node: it has three independent roots for a theorem-backed local occurrence family, actual active support, and canonical minimal representation. Four independent physical registries remain missing. In parallel, the determinant-two core now has a fixed-X-power all-prefix packet theorem for Lebesgue-almost every fixed phase. That theorem does not reach a named physical phase.

MVP7 turns this state into a dynamic classifier. Complete architecture routes and arithmetic method subroutes inhabit disjoint registries. Only a stopped architecture can trigger rerouting, and only through a fresh registry and a source-backed typed crosswalk. The uncontrolled metric-to-atomic nonimplication stops one arithmetic method cell; it does not stop the direct fixed-phase twist or the selected occurrence architecture. A six-axis projection records all remaining quantifier changes.

After source, schema, type, route, frontier and Endpoint V4 validation, the current verdict is

$$\mathsf{NOT\_TESTABLE}.$$

The selected first missing object is the source-backed local occurrence-edge family, while the full seven-node blocker antichain is preserved. The arithmetic advance remains L1 actual-core phase metric, not program-positive L2. No strict 1/400, prime-pair lower bound or twin-prime theorem is claimed.

1 Imported state

Let $\mathcal{M}_{171}$ be the canonical TPC-171 manifest. MVP7 locks its generator, manifest, audit, schemas and paper source, and also locks the two exact scoped-stop sources used by the classifier. Every hash has integrity semantics only.

The selected map root has the minimal not-testable antichain

$$\begin{aligned}\mathcal{B}_{\mathrm{NT}} = \{&\mathsf{H1.source\_backed\_local\_occurrence\_edge\_family},\\ &\mathsf{H1.actual\_active\_support\_certificate},\\ &\mathsf{H1.canonical\_minimal\_representation\_certificate},\\ &\mathsf{H9.literal\_weight\_registry}, \mathsf{H9.phase\_cell\_registry},\\ &\mathsf{H9.endpoint\_registry}, \mathsf{H9.normalization\_registry}\}.\end{aligned} \tag{1}$$

The first element is a reporting representative selected by the declared obligation order. It does not replace the other six.

The separate parent-ready frontier is

$$\mathcal{F}_{\mathrm{open}} = \{\mathsf{O161.bad\_endpoint\_pointwise\_fixed\_atom}, \mathsf{O161.direct\_additive\_twist\_fixed\_atom}\}. \tag{2}$$

The sets in (1) and (2) are disjoint in both status and role. A missing artifact and an open theorem target are not interchangeable.

On the arithmetic side the strongest imported result is

$$\begin{gathered}\mathsf{A170.metric\_packet\_corridor}\\ = \mathsf{PROVED\_L1\_ACTUAL\_CORE\_PHASE\_METRIC\_}\\ \mathsf{PACKET\_CORRIDOR}.\end{gathered}\tag{3}$$

It is maximal over every prefix in each declared terminal shell and, on the prescribed packet schedule, holds for Lebesgue-almost every fixed phase with every exponent $\delta < 1/4$. Its analytic norm is $L^2_{\mathrm{phase,maximal,BC}}$. The program-positive L2 flag, named-fixed-atom flag and production-phase-registry flag are all false.

2 Six-axis projection

For each theorem the TPC-171 contract records the signature

$$\mathfrak{q} = (\mathfrak{c}, \mathfrak{p}, \mathfrak{e}, \mathfrak{s}, \mathfrak{d}, \mathfrak{a}),\tag{4}$$

with carrier, phase, endpoint, scale, decay and support coordinates. MVP7 displays the strongest proved signature and the physical signature required at the endpoint:

axis	proved	required
$\mathfrak{c}$	explicit packet corridor	actual fixed-h_0 packet
$\mathfrak{p}$	Lebesgue-a.e. fixed phase	named fixed atom
$\mathfrak{e}$	all prefixes in a θ-shell	deterministic all-prefix
$\mathfrak{s}$	eventually on prescribed schedule	deterministic all-scale
$\mathfrak{d}$	fixed-X phase-metric power	fixed-X fixed-atom power
$\mathfrak{a}$	actual arithmetic core	actual active support.

(5)

No coordinate in (5) has yet been bridged. In particular, a fixed-X exponent in the fifth row cannot pay for missing phase or support quantifiers in the second and sixth rows.

Proposition 2.1 (Atomic firewall)**.** *The TPC-170 nonimplication stops only the method*

$$\mathsf{phase\_metric\_uncontrolled\_atomic}.\tag{6}$$

It does not stop a source-backed metric-to-fixed-atom bridge, a pointwise direct-twist theorem, a pointwise bad-endpoint theorem or an architecture route.

Proof. The nonimplication uses an arbitrary atomic measure concentrated on the schedule-dependent metric null set. Its exact scope is uncontrolled atomic promotion. A source-backed bridge would add new information about the physical phase, and a pointwise theorem would replace the metric quantifier. Neither lies in the stopped cell. Architecture routes concern the construction and return of the physical carrier and have a different route type. □

3 Two route families

The classifier contains two registries.

Architecture routes. These are complete candidate paths to the physical endpoint:

$$\mathcal{R}_{\mathrm{arch}} = \{\mathsf{current\ archive}, \mathsf{occurrence\ map}{+}, \mathsf{scalar} + \mathsf{ETO}\}.\tag{7}$$

The current-archive-only cell is stopped in the exact scope CURRENT_ARCHIVE_FIELDS_ONLY. The selected occurrence-augmented map and alternative scalar-plus-ETO architectures remain NOT-TESTABLE. Their universe is not proved complete.

Arithmetic method subroutes. These are theorem choices inside an architecture:

$$\begin{array}{ccc}\textsf{periodic major}, & \textsf{direct twist fixed atom}, & \textsf{phase metric source backed}, \\ \textsf{phase metric uncontrolled atomic}, & \textsf{bad endpoint pointwise fixed atom}. & \end{array} \tag{8}$$

No element of (8) is architecture-reroute eligible.

Definition 3.1 (Legal architecture reroute)**.** A reroute is legal only if:

(i) the selected complete architecture has an exact source-backed stop in matching scope;

(ii) the alternative is another architecture route, not an arithmetic method;

(iii) the alternative has a genuinely fresh registry; and

(iv) a source-backed typed crosswalk connects the old and new carriers, scopes and normalizations.

Renaming the stopped registry, citing an arithmetic method or submitting an unregistered theorem label makes the snapshot INVALID, not REROUTE.

4 Evidence modes

The production classifier runs in SOURCE_LOCKED mode. Every stop, route-universe theorem and reroute crosswalk must resolve to a canonical repository path, hash, claim type, route type and exact scope. The current production registry contains:

1. the current-archive architecture stop from TPC-154; and
2. the uncontrolled-atomic arithmetic-method stop from TPC-170.

The second record cannot be consumed as architecture evidence.

Classifier reachability is tested in the disjoint SYNTHETIC_REACHABILITY mode. Its records carry

$$\textsf{NONE\_SYNTHETIC\_REACHABILITY\_ONLY} \tag{9}$$

semantics and have no file path or theorem hash. They prove that the finite classifier logic reaches each verdict; they prove no mathematical premise. Moving one of them into production mode is a validation failure.

5 Endpoint V4 classifier state

Endpoint Ledger V4 retains the six literal gates:

V4.1 literal physical coefficients;

V4.2 fixed physical $h_0 = 2$;

V4.3 physical atomic normalization;

V4.4 canonical or minimal representation;

V4.5 actual active support;

V4.6 strict 1/400 endpoint slack.

Only V4.2 is currently proved. The four physical registries and all other literal gates remain NOT-TESTABLE.

Each H9 leaf is a data registry and has decay coordinate NONE. The phase leaf may identify a named phase and the endpoint leaf may identify deterministic endpoints, but neither fact is an estimate.

A fixed-X decay coordinate may appear only on an independent arithmetic theorem that consumes these records.

The phase-metric charge records

$$\delta < \frac{1}{4} \quad \text{on a Lebesgue-a.e. explicit packet schedule.} \tag{10}$$

It is explicitly ineligible for the named-fixed-atom ledger. Therefore

$$\sigma_{\text{named atom}} = 0, \qquad \sigma_{\text{required}} = \frac{1}{400}. \tag{11}$$

The physical loss and strict net slack are undefined. Unknown quantities are not zero.

Definition 5.1 (Arithmetic-frontier state)**.** The result ARITHMETIC-FRONTIER is available only when the complete structural carrier, production physical registry, named fixed phase and deterministic physical endpoint are all proved, and every remaining unresolved node is a scope-matched parent-ready positive-L2 arithmetic target.

Thus the current phase-metric theorem cannot itself trigger ARITHMETIC-FRONTIER. It leaves structural and physical artifacts unavailable and does not have the named-atom quantifier.

6 Ordered classifier

Validation occurs before mathematical classification. Failure of a source lock, schema, route type, evidence type, frontier invariant or ledger identity returns the outer result INVALID.

Definition 6.1 (Ordered valid-snapshot verdicts)**.** For a valid snapshot, the first true predicate in

$$\boxed{\begin{array}{c}\text{GO}, \quad \text{ARCHITECTURE-INFEASIBLE}, \quad \text{REROUTE}, \quad \text{STOP-ROUTE}, \\ \text{NOT-TESTABLE}, \quad \text{ARITHMETIC-FRONTIER}, \quad \text{OPEN}\end{array}} \tag{12}$$

is returned.

(i) GO: every active requirement is proved in matching scope, all literal gates pass, and the exact endpoint slack is strictly positive.

(ii) ARCHITECTURE-INFEASIBLE: an independent source-backed theorem proves the architecture universe complete and every complete architecture cell has an exact stop.

(iii) REROUTE: the selected architecture is stopped and a legal alternative satisfies definition 3.1.

(iv) STOP-ROUTE: the selected architecture is stopped and no legal alternative is selected.

(v) NOT-TESTABLE: the selected-root typed blocker antichain is nonempty.

(vi) ARITHMETIC-FRONTIER: the conditions of definition 5.1 hold.

(vii) OPEN: every other valid state.

Theorem 6.2 (Classifier totality and reachability)**.** *Validation followed by* (12) *returns exactly one of*

$$\begin{array}{c}\text{INVALID}, \quad \text{GO}, \quad \text{ARCHITECTURE-INFEASIBLE}, \quad \text{REROUTE}, \\ \text{STOP-ROUTE}, \quad \text{NOT-TESTABLE}, \quad \text{ARITHMETIC-FRONTIER}, \quad \text{OPEN}.\end{array}$$

Every result is reached by a finite typed regression state.

Proof. Validation is deterministic and either fails or produces a valid state. On a valid state the ordered list has a final catch-all, so a unique first true predicate exists. The synthetic suite supplies one separate assumed-predicate state for each valid verdict and one malformed route-type state for INVALID. The evidence-mode firewall (9) prevents these reachability witnesses from entering the production result. □

7 MVP7 decision

Theorem 7.1 (Source-locked MVP7 verdict)**.** *For the TPC-163–171 source-locked state,*

$$\boxed{\begin{array}{c}\mathrm{Verdict}_{163:171} = \textsc{not-testable},\\ \mathsf{FirstMissing} = \mathsf{H1.source\_backed\_local\_}\\ \mathsf{occurrence\_edge\_family}.\end{array}} \tag{13}$$

The complete blocker antichain is (1)*; the two pointwise arithmetic targets* (2) *remain parent-ready* OPEN.

Proof. All production source locks, route types, evidence types, frontier lists and Endpoint V4 identities validate, excluding INVALID. The selected occurrence map architecture has no source-backed stop, and architecture-universe completeness is unproved. Hence neither ARCHITECTURE-INFEASIBLE, REROUTE nor STOP-ROUTE applies. The nonempty selected-root antichain (1) then triggers NOT-TESTABLE before ARITHMETIC-FRONTIER or OPEN.

Independently, the phase projection (5) is incomplete on every axis, the named-atom exponent in (11) is zero and five literal gates remain unknown. Thus GO is also false. The declared obligation order chooses the first element of (1) without discarding the other minimal blockers. □

Corollary 7.2 (Exact progress label)**.** *The current state has:*

1. L1 *structural localization into three production roots;*
2. L1 *actual-core phase-metric all-prefix packet control;*
3. *no actual active-support certificate;*
4. *no named-fixed-phase theorem; and*
5. *no program-positive* L2 *or strict* $1/400$ *endpoint.*

8 Mutation audit and next decisions

The audit regenerates the snapshot dynamically from TPC-171. It rejects source-hash drift and proof semantics; a Lebesgue-a.e. phase promoted to a named atom; a metric exponent promoted to program-L2; a metric charge entered as fixed-atom power; an arithmetic method inserted into the architecture universe; an arithmetic method marked reroute eligible; route-universe completeness without evidence; a fabricated stop label; the uncontrolled-atomic stop promoted to a direct-twist stop; a collapsed blocker antichain; an OPEN/NOT-TESTABLE frontier merge; a positive named-atom exponent without a bridge; and a $1/400$ pass without the literal gates.

The next structural construction must supply a theorem-backed local occurrence-edge family, with source locators and compatible overlaps. Actual active support and canonical minimal representation remain parallel obligations, not consequences. The arithmetic branch has two live choices: prove the production named phase lies in the TPC-170 good set through a source-backed bridge, or establish a pointwise fixed-phase theorem. If one choice fails in its exact method cell, the other remains open.

9 Conclusion

MVP7 confirms that the route is still live but not yet testable on the physical carrier. The new arithmetic corridor is real and stronger than the earlier almost-endpoint theorem: it reaches all prefixes with a fixed-X metric exponent along an explicit schedule. The remaining gap is now a sharp quantifier interface, not an unspecified "phase problem." Until a named physical atom and the three structural roots are supplied, the only source-compatible verdict is NOT-TESTABLE. This statement neither implies nor suggests a prime-pair lower bound or the twin-prime conjecture.

References

[1] Rick Durrett. *Probability: Theory and Examples.* Cambridge University Press, fifth edition, 2019.

[2] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory.* Cambridge University Press, 2007.

[3] Liang Wang. A Conservative Completion-Fiber Obstruction for the Cut-to-Occurrence Lift, 2026. TPC-154 manuscript.

[4] Liang Wang. A Refined H1 Crosswalk Frontier: Three Independent Production Roots, 2026. TPC-166 manuscript.

[5] Liang Wang. Metric Packet-Corridor Return: Borel–Cantelli, Abel Summation, and the Fixed-Atom Stop, 2026. TPC-170 manuscript.

[6] Liang Wang. Source-Locked Occurrence–Phase Return Integration: Three Structural Roots, Metric All-Prefix Control, and the Named-Atom Firewall, 2026. TPC-171 manuscript.